Modeling linear and non-linear drag in horizontal oscillatory motion

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Exponentially damped oscillatory motion is often encountered early in a physics student's education. The closed form solution in the case of linear-in-velocity drag makes this an attractive first model, but realistic systems typically include constant- and/or quadratic-in-velocity drag. The equations of motion for non-linear drag can be solved numerically so that data can be fit using more accurate models. A data set is presented and analyzed which students can use to explore topics such as numerical modeling techniques, evaluating and quantifying the agreement between a model and data, and the behavior of friction. As an example, we aim to determine whether linear or quadratic drag better models the motion of a cart. Code and data are provided for physics instructors' use.

I. INTRODUCTION

Treatment of the damped harmonic oscillator with linear drag is ubiquitous in introductory physics textbooks (e.g., [3]). The relatively simple, closed-form solution for the motion in this case makes this an attractive first model. Some physics education articles have presented constant drag and the resulting linearly decaying oscillation amplitude [4, 7] or differences in motion resulting from linear vs constant drag [12, 14]. In order to keep the presentations accessible to students, constant drag motion is generally studied from an energy conservation perspective. In this way, the oscillation's envelope can be determined without needing to determining the full motion of the system. Other subtleties of the energy of damped oscillating systems have also been considered [1, 5]. Quadratic drag is typically only discussed relatively superficially [8–11]. Even in advanced texts, non-linear drag is only considered in the abstract [2, 6].

Here, we aim to provide a full description of damped oscillatory motion, including any combination of constant, linear, and/or quadratic drag. Doing so is illuminating not only in terms of physical modeling, but can also as a way to expose students to more general fitting practices [15] and important computational tools such as Runge-Kutta integration and gradient descent minimization [13]. We outline our data collection and fitting methods in Section II, interpret the results of our fits in Section III, and discuss our conclusions and provide the code and data in Section IV.

II. DATA COLLECTION AND FITTING METHODS

A cart on a track was attached via springs to posts on either side. The track was clamped to a sturdy bench which is built into the wall to minimize motion of the track, etc. The springs were fixed to the posts using rubber bands in order to minimize their movement. A block was attached to the top of the cart to lower its oscillation frequency. Position vs time data were collected using the Vernier motion encoder system and LoggerPro. See Fig. 1.

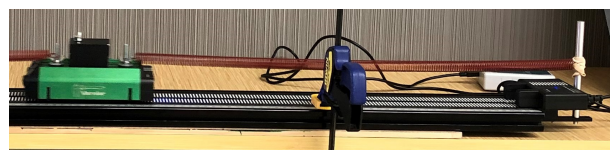

FIG. 1. (Color online.) Experimental setup. The spring and clamp on the left side are set up similarly.

By collecting spring force vs displacement data in LoggerPro and fitting to a line, we determined the effective spring constant of the system to be (48.84 ± 0.06) N/m. The mass of our cart/block system is (1.12 ± 0.01) kg. Uncertainties represent one standard deviation.

We model the motion of our system using Newton's second law with a spring force described by Hooke's law and various candidate models of drag

$$m\ddot{x} = -k\left(x - x_{\mathrm{eq}}\right) - c_1\dot{x} - \left(c_0 + c_n\left|\frac{\dot{x}}{1\ \mathrm{m/s}}\right|^n + c_2\dot{x}^2\right)\mathrm{sign}\left(\dot{x}\right), \tag{2.1}$$

where the sign function

$$\mathrm{sign}\left(v\right) = \begin{cases} 1 & v > 0 \\ -1 & v < 0 \\ 0 & v = 0 \end{cases} \tag{2.2}$$

ensures that drag always oppose motion. We have included the possibility of a term with unknown ve-

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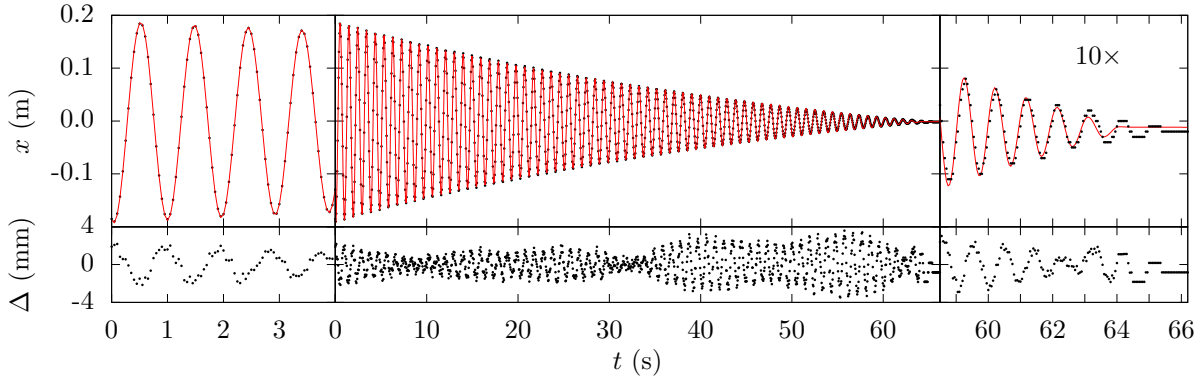

FIG. 2. (Color online.) Fit for model CON. Top: data (black points) and fit (red line); bottom: residuals $\Delta_i = x_i - f(t_i; \vec{A})$; center: full duration; left and right: focus on early and late times. The vertical axis of the top right plot has been zoomed in by an order of magnitude to better show the data.

locity dependence to see if fitting indicates a preference for linear or quadratic drag. There is no easy to work with, closed-form solution to (2.1) when any of c_0, c_2, and c_n are non-zero; in order to explore these more mathematically difficult models, we resort to numerical integration (here, using classic fourth-order Runge-Kutta).

Our data set consists of N data points (t_i, x_i). Our numerical integration yields predictions of the cart's positions according to (2.1), the solution to which we call $f(t; \vec{A})$. The fit parameter vector $\vec{A}$ contains initial data for (2.1) along with parameters describing the system

$$\vec{A} = \langle x_0, v_0, m, k, x_{\text{eq}}, \ldots \rangle \,; \tag{2.3}$$

note that the number of parameters in $\vec{A}$ is different for different models of drag. We quantify the goodness of a fit using

$$\chi^2 = \sum_{i=1}^{N} \Big(x_i - f(t_i; \vec{A}) \Big)^2 . \tag{2.4}$$

Gradient descent then attempts to find fit parameters $\vec{A}$ which minimize χ^2. Further details of the implementation of these methods can be found in documentation accompanying the code (see the end of Section IV).

Note that not all the fit parameters are independent; (2.1) can be divided by an overall factor to effectively remove one degree of freedom from the fit. In particular, we could divide through by k, fit for m/k, c_1/k, etc. In practice, we will accomplish this by simply fixing the measured value of k quoted above in the code and not including the component of the gradient in the k direction. The interdependence of the fit parameters is further reflected in the exponentially damped case, where the analytic solution only includes five degrees of freedom though the model contains six parameters (see [16]).

III. ANALYSIS

We fit the position vs time data using five different models, named C1, C01, C02, C0N, and C012N. We include a model with only linear drag (C1) and also a model where all forms of friction listed in (2.1) are present (C012N). These models may lead to under- and over-fitting, respectively, but give us benchmarks for the worst and best we might expect from our models based on the methods we are using. We expect rolling friction to be present and therefore include constant drag in all three of the realistic models; these models are differentiated by having linear (C01), quadratic (C02), or v^n (C0N) drag in addition to constant drag. In models containing v^n terms, we initialize fitting with a guess of $n = 3/2$, halfway between the linear and quadratic models. Fig. 2 shows the data along with the C0N fit and residuals. Fig. 3 and Table I display residuals and parameters for all five models.

We see in Fig. 2 that, early on, the residuals oscillate with the same frequency as the position data and that at late time the fit is essentially constant so that $\Delta_i \approx x_i$. Together, these observations suggest that x_0, v_0, m, and k are relatively well determined and the envelope – as determined by the drag – is the main source of the fit's error. Similar observations were made for all models and the example shown in Fig. 2 appears to be representative.

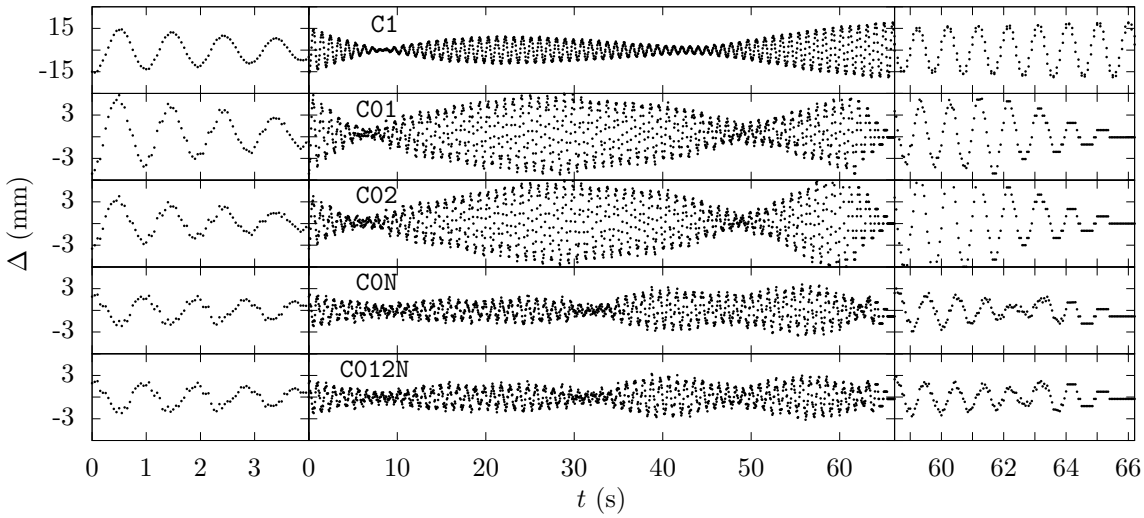

FIG. 3. Residuals for five different models of drag. Left, right, and center are as in Fig. 2; each row of graphs corresponds to a column in Table I and is labeled with the c values which are non-zero in that particular model. Top to bottom: `C1`, `C01`, `C02`, `C0N`, and `C012N`. Note that the residuals for `C1` have a different vertical scale.

IV. CONCLUSIONS

Estimating relevant parameters (see [17]), we estimate the Reynolds number of the cart moving through air to be ~ 2000, which is near the transition between laminar (linear drag) and turbulent flow (quadratic drag). The χ^2 value for `C01` is slightly smaller than the χ^2 value for `C02`, though both χ^2 values are close to the best we would expect given the results of `C012N`. When the model's velocity dependence was allowed to vary, the fitting algorithm did not move the power appreciably from the initial guess of 3/2. We conclude that, for much of its motion, the cart is near the critical Reynolds number so that neither linear nor quadratic drag provides a complete model.

Another interesting observation is that the fit values for the mass of the cart were all larger than the measured value [18]. This makes sense, given that the springs are not massless, though our model (2.1) assumes they are. This is why we used the measured value of k in our fits instead of the measured value of m.

In addition to being interesting in its own right, this project raises a number of pedagogical points which may be useful in physics education. The code and data can be found at `https://github.com/kyle-slinker/teaching-resources`, and physics teachers are encouraged to use them in their classrooms. A wide range of projects should be possible, based on students' preparations: given the Python code as a black-box, students could fit data they take themselves; the effects of different types of drag can be studied; in a computational physics class, analytic solutions for the $c_0 = c_2 = c_n = 0$ case can be compared to numerical solutions. Or, these materials might inspire a more in-depth long-term project; the results presented here, for example, were obtained as part of a research project carried out over multiple trimesters by an experienced physics student.

TABLE I. Fit parameters for the five models whose residuals are shown in Fig. 3; data columns here correspond to rows there. Spring constant values were fixed in the fits. Significands are given separately from magnitudes; magnitudes are given along with units, to the right. See also note [16].

	`C1`	`C01`	`C02`	`C0N`	`C012N`		
k	48.84	48.84	48.84	48.84	48.84		N/m
x_0	−2.000	−1.800	−1.815	−1.869	−1.869	$\times 10^{-1}$	m
v_0	−3.253	−3.037	−3.051	−3.051	−3.052	$\times 10^{-1}$	m/s
m	1.147	1.147	1.146	1.147	1.147		kg
x_{eq}	−1.604	−2.059	−1.806	−1.588	−1.574	$\times 10^{-3}$	m
c_0	–	2.601	3.057	2.243	2.209	$\times 10^{-2}$	N
c_1	83.12	21.33	–	–	7.436	$\times 10^{-3}$	kg/s
c_2	–	–	20.31	–	5.139	$\times 10^{-3}$	kg/m
c_n	–	–	–	4.033	2.822	$\times 10^{-2}$	N
n	–	–	–	1.500	1.500		–
χ^2	56.74	8.700	9.722	2.229	1.945	$\times 10^{-3}$	m^2

V. ACKNOWLEDGEMENTS

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[16] The values shown in the leftmost numerical column of Table I were determined using LoggerPro's fit to an exponentially damped harmonic oscillator. The equation $x(t) = Ae^{-Bt}\sin(Ct + D) + E$ yields five fit values

$$A = 0.2048 \text{ m} \qquad D = 4.461 \text{ rad}$$
$$B = 0.03624/\text{s} \qquad E = -0.001604 \text{ m}$$
$$C = 6.526 \text{ rad/s},$$

which can be related algebraically to the system parameters. Our fitting code was able to determine a fit in terms of the system parameters directly with a χ^2 value which did not differ significantly from the one found with the LoggerPro fit.

[17] Using approximate values for the mean speed of the cart during its first full oscillation, the area of a face of the cart, and the viscosity of air, we find

$$\text{Re} = \frac{v\sqrt{A}}{\nu} \approx \frac{(0.7 \text{ m/s})\sqrt{20 \text{ cm}^2}}{1.4 \times 10^{-5} \text{ m}^2/\text{s}} = 2200.$$

Given the rough nature of the calculation, 2000 is likely helpful as an order of magnitude estimate at best.

[18] Here we determine statistical significance based on the uncertainty of the measured values. It is possible to determine variance and covariance for the fit values, but this is likely only feasible for much more advanced students.